Cosmoserve to start space debris removal



Telangana's Minister for IT, Electronics and Communications, D Sridhar Babu (*sixth from right*), with the Cosmoserve Space team at its newly inaugurated facility in Hyderabad on Thursday PHOTO: COSMOSERVE SPACE

SHINE JACOB
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Cosmoserve Space, an Indian startup developing active debris removal (ADR) technologies, unveiled Base 1, India's first private and the world's sixth facility equipped with space vision effect and microgravity simulation, in Hyderabad on Thursday.

Beyond removing space debris, it will support satellite servicing, including in-orbit repairs and refuelling.

Cosmoserve also announced a $6 million seed funding round, taking its total funding to over $9 million. Japan-based Rebright Partners joined Indian investors Aranya Holdings, Naandi Ventures, AUM Ventures and Shakti VC in the round.

In July 2026, Cosmoserve flew its first payload, Mission Embrace, aboard Skyroot Aerospace's Vikram-1 rocket. It is planning to come up with the next technology demonstration mission by mid-2027 and its first commercial mission in early 2028.

This will be part of the larger road map by the Telangana government to transform Hyderabad into one of the most affordable space hubs in the world. The state is also actively planning an exclusive Space City within the upcoming Bharat Future City.

"Facilities like Cosmoserve's Base 1 show exactly why Hyderabad is leading the country in space innovation. We are now home to over 30 spacetech companies, including India's first spacetech unicorn and a few more soonicorns (soon to be unicorns). They are building everything from launch vehicles and space data centres to debris removal systems and drug discovery platforms," said D Sridhar Babu, Minister for Information Technology, Electronics and Communications, Telangana.

The facility brings full-stack space servicing capabilities under one roof.

Base 1 houses four key facilities: An Orbital Rendezvous and Capture Lab to simulate spacecraft capture operations; a Space Robotics Innovation Lab to develop robotic capture systems; an ISO Class 8 clean room for satellite integration; and a Mission Simulation and Control Centre to manage upcoming missions. "We will start commercial operations by mid-2028, and this will open our doors to a global market projected to reach $14.5 billion by 2030," added Chiranjeevi Phanindra, founder and chief executive officer (CEO) of Cosmoserve Space.

The funding will support Base 1, technology demonstrations and two upcoming missions. Space debris is one of the biggest challenges facing the global space industry. Of the around 28,000 satellites in orbit, nearly 10,000 are dead or non-operational.

Earth's orbit is crowded with an estimated 68,000 debris objects smaller than 10 cm, 1.5 million fragments measuring 1-2 cm, and nearly 230 million tiny pieces ranging from 1 millimetre (mm) to 1 cm.

Even fragments just a few mm wide can pose serious risks to operational satellites, spacecraft and missions.

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